In [1]:
```
#Practice Program-6
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```

In [2]:
```
"""1.Draw a histogram showing distribution of flipper_length across species for penguin dataset
2.Create a stack plot and step plot- – Write down your interpretations
3.Draw a bivariant KDE plot to show relationship between horsepower and displacement [use mpg dataset] – Write do
4.Draw a correlation heatmap for diamonds dataset, add annotation, change colour pallet and write your interpreta
5.Draw a scatter plot to visualizing relationships between flower measurements. [e.g. sepal_length, petal_width]
Write your interpretations"""
```

In [3]:
```
import seaborn as sns
```

In [4]:
```
sns.get_dataset_names()
```

Out[4]:
```
['anagrams',
 'anscombe',
 'attention',
 'brain_networks',
 'car_crashes',
 'diamonds',
 'dots',
 'dowjones',
 'exercise',
 'flights',
 'fmri',
 'geyser',
 'glue',
 'healthexp',
 'iris',
 'mpg',
 'penguins',
 'planets',
 'seaice',
 'taxis',
 'tips',
 'titanic']
```

In [5]:
```
df = sns.load_dataset("penguins")
df
```

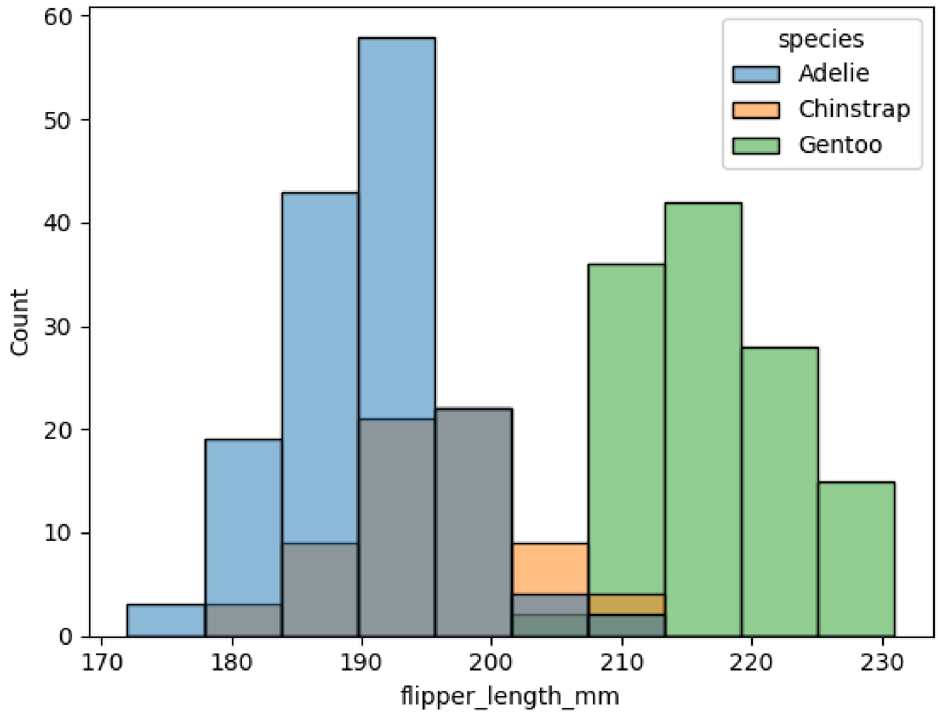
Out[5]:

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	Male
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	Female
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	Female
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0	Female
...	...	...	...	...	...	...	...
339	Gentoo	Biscoe	NaN	NaN	NaN	NaN	NaN
340	Gentoo	Biscoe	46.8	14.3	215.0	4850.0	Female
341	Gentoo	Biscoe	50.4	15.7	222.0	5750.0	Male
342	Gentoo	Biscoe	45.2	14.8	212.0	5200.0	Female
343	Gentoo	Biscoe	49.9	16.1	213.0	5400.0	Male

344 rows × 7 columns

In [6]:
```
# 1.Draw a histogram showing distribution of flipper_length across species for penguin dataset
sns.histplot(df,x="flipper_length_mm",hue ="species")
```

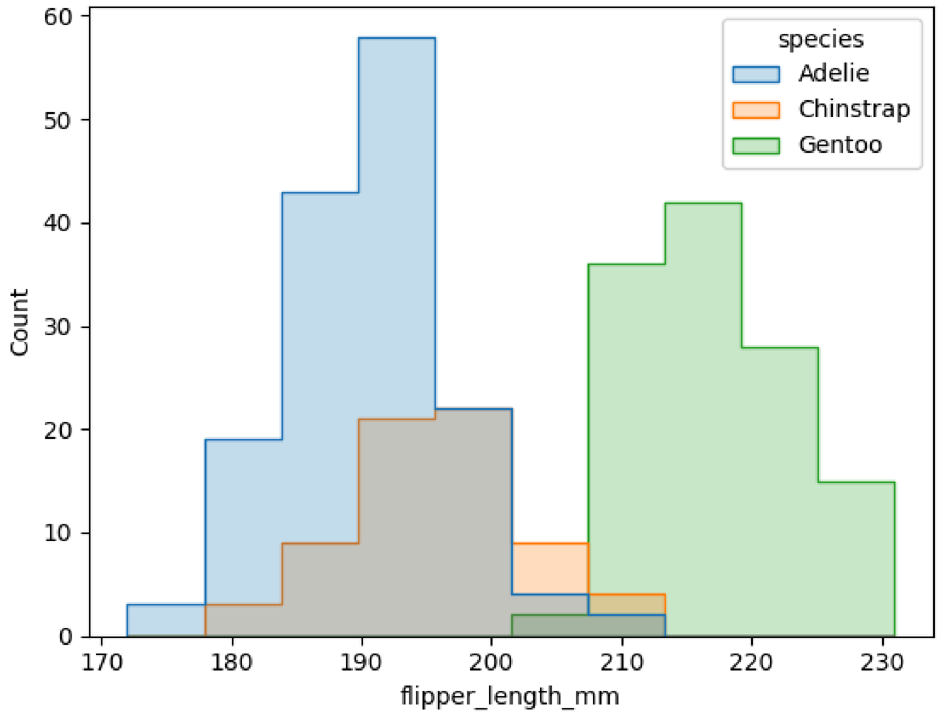
Out[6]: <Axes: xlabel='flipper_length_mm', ylabel='Count'>



In [7]:
```
"""---Interpretation---
1.There is a clear separation between species. Adelie and Chinstrap penguins have shorter flippers, while Gentoo
2.Adelie mostly falling between 185mm and 195mm.
3.Chinstrap group overlaps heavily with the Adelie species. They appear mostly in the 190mm to 200mm range.
4.Gentoo are the largest, There is almost no overlap between Gentoo and Adelie penguins.
"""
```

In [8]:
```
# 2.Create a stack plot and step plot- - Write down your interpretations
sns.histplot(df,x="flipper_length_mm",hue ="species",element ="step")
```

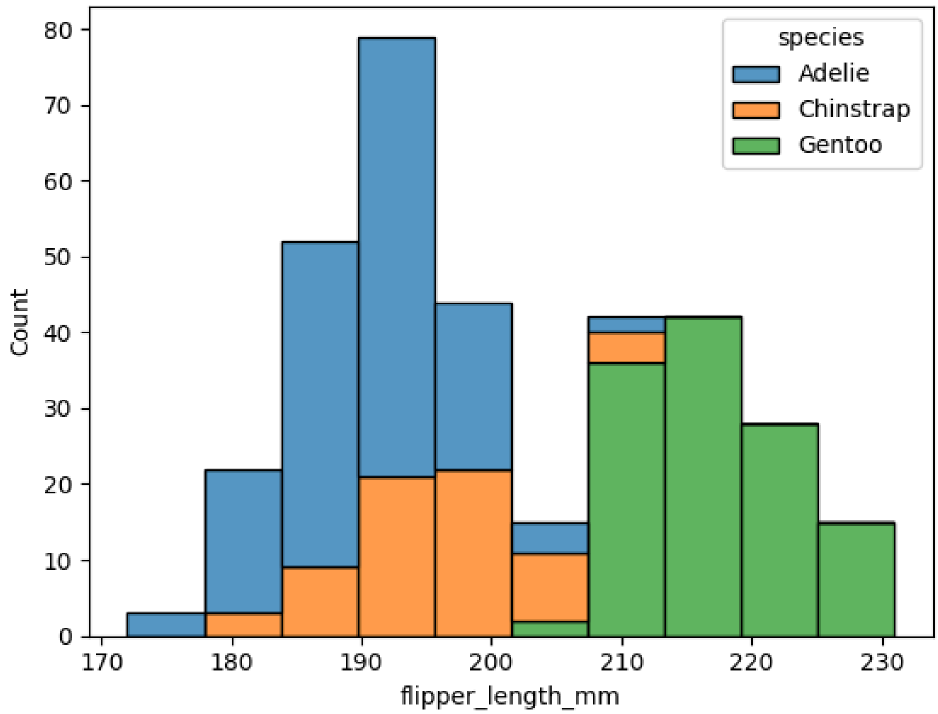
Out[8]: <Axes: xlabel='flipper_length_mm', ylabel='Count'>



In [9]:
```
"""----Interpretation----
1.The data is bimodal.
2.The Gentoo species is almost entirely separated from the others, making flipper length a strong predictor for i
3.While Adelie and Chinstrap overlap, Chinstrap penguins tend to have slightly longer flippers on average than Ad
```

In [10]:
```
sns.histplot(df,x="flipper_length_mm",hue ="species",multiple ="stack")
```

Out[10]: <Axes: xlabel='flipper_length_mm', ylabel='Count'>



In [11]:
```
"""---Interpretation---
1.The total height of the bars shows the cumulative count of all penguins.
2.The highest concentration of penguins is in the 190–195mm range.
3.It clearly shows that Gentoo penguins account for almost all observations above 210mm, confirming they are the
```

In [12]:
```
# 3.Draw a bivariant KDE plot to show relationship between horsepower and displacement [use mpg dataset] - Write
df1 = sns.load_dataset("mpg")
df1
```

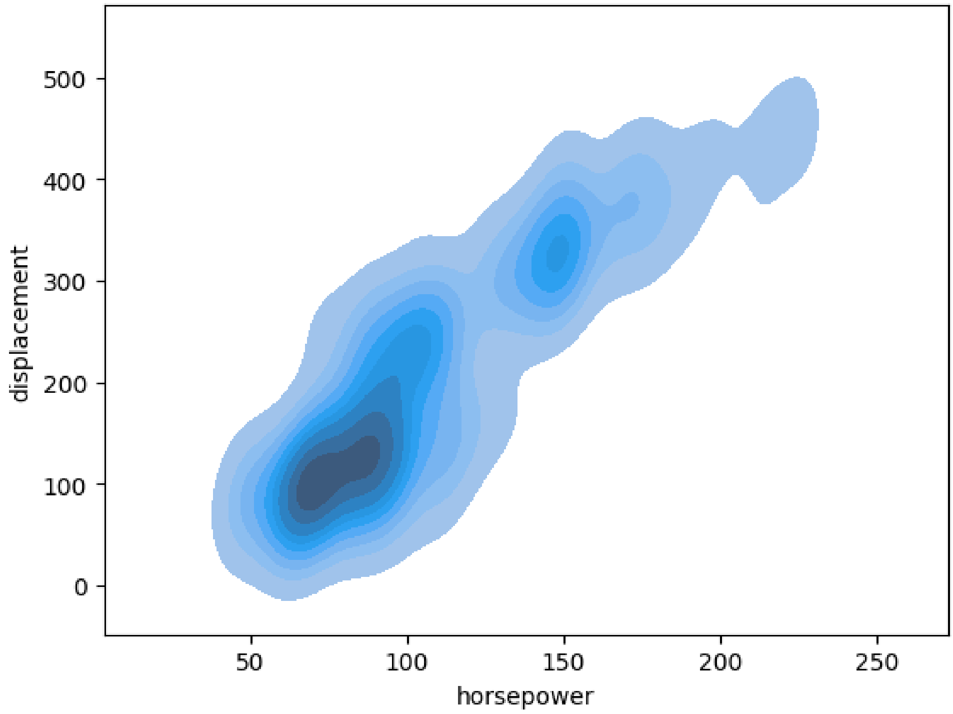
Out[12]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino
...	...	...	...	...	...	...	...	...	...
393	27.0	4	140.0	86.0	2790	15.6	82	usa	ford mustang gl
394	44.0	4	97.0	52.0	2130	24.6	82	europe	vw pickup
395	32.0	4	135.0	84.0	2295	11.6	82	usa	dodge rampage
396	28.0	4	120.0	79.0	2625	18.6	82	usa	ford ranger
397	31.0	4	119.0	82.0	2720	19.4	82	usa	chevy s-10

398 rows × 9 columns

In [13]:
```
sns.kdeplot(df1,x="horsepower",y="displacement",fill=True)
```

Out[13]: <Axes: xlabel='horsepower', ylabel='displacement'>



In [14]:
```
"""---Interpretation---
1.Most vehicles are concentrated in the 70-100 hp and 100 displacement range.
2.The hotspots indicate three distinct classes of engines (small, mid-range, and high-performance).
3.The data is tighter at the low end but spreads out at the high end, showing that larger engines have more varie
"""
```

In [15]:
```
# 4.Draw a correlation heatmap for diamonds dataset, add annotation, change colour pallet and write your interpre
df2=sns.load_dataset("diamonds")
df2
```

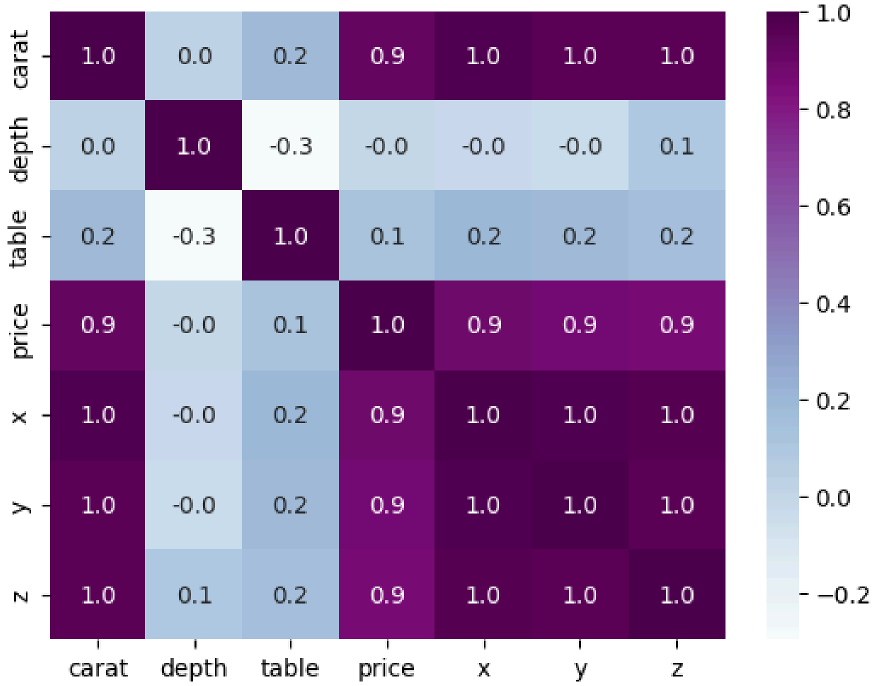
Out[15]:

	carat	cut	color	clarity	depth	table	price	x	y	z
0	0.23	Ideal	E	SI2	61.5	55.0	326	3.95	3.98	2.43
1	0.21	Premium	E	SI1	59.8	61.0	326	3.89	3.84	2.31
2	0.23	Good	E	VS1	56.9	65.0	327	4.05	4.07	2.31
3	0.29	Premium	I	VS2	62.4	58.0	334	4.20	4.23	2.63
4	0.31	Good	J	SI2	63.3	58.0	335	4.34	4.35	2.75
...	...	...	...	...	...	...	...	...	...	...
53935	0.72	Ideal	D	SI1	60.8	57.0	2757	5.75	5.76	3.50
53936	0.72	Good	D	SI1	63.1	55.0	2757	5.69	5.75	3.61
53937	0.70	Very Good	D	SI1	62.8	60.0	2757	5.66	5.68	3.56
53938	0.86	Premium	H	SI2	61.0	58.0	2757	6.15	6.12	3.74
53939	0.75	Ideal	D	SI2	62.2	55.0	2757	5.83	5.87	3.64

53940 rows × 10 columns

In [16]:
```
sns.heatmap(df2.corr(numeric_only = True),cmap='BuPu',annot=True ,fmt=".1f")
```

Out[16]:
```
<Axes: >
```



In [17]:
```
"""---Interpretation---
1.Carat, x, y, and z all have a strong positive correlation with price. This means as a diamond gets bigger or he
2.Carat and the dimensions (x, y, z) are perfectly correlated.
3.Features like depth and table show almost zero correlation with price. Their proportions don't significantly di
"""
```

In [18]:
```
# 5.Draw a scatter plot to visualizing relationships between flower measurements. [e.g. sepal_length, petal_width
# Write your interpretations
df3 = sns.load_dataset('iris')
df3
```

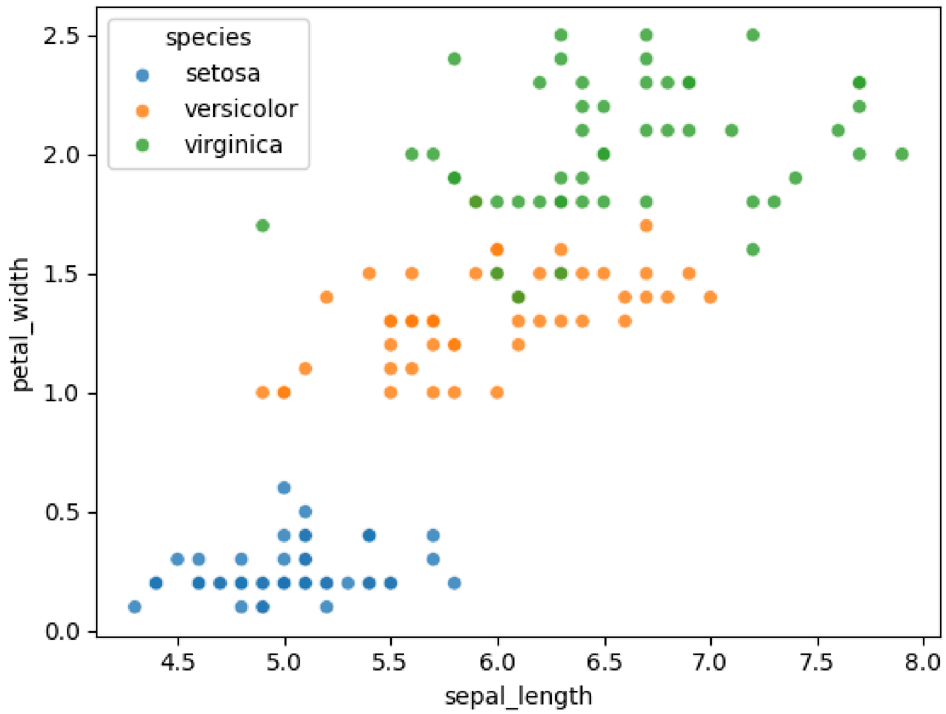
Out[18]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

In [19]:
```
sns.scatterplot(df3,x="sepal_length" ,y="petal_width",hue ="species",alpha =0.8)
```

Out[19]: <Axes: xlabel='sepal_length', ylabel='petal_width'>



In [20]:
```
# """---Interpretation---
# 1.The Setosa species is clearly isolated with the smallest dimensions.
# 2.Versicolor and Virginica show a positive correlation between length and width, but they overlap in the middle
# """
```

In [ ]:
```
```